

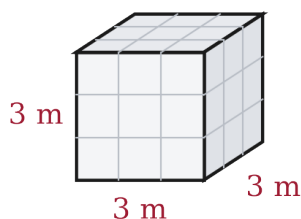
Cube ValuesAdditional Practice 7A.02.03 **ANSWER KEY**

1. Complete the table. Each volume is the side length multiplied by itself three times.

Side s	1	2	3	4	5
Volume s^3	1	8	27	64	125
Side $-s$	-1	-2	-3	-4	-5
Volume $(-s)^3$	-1	-8	-27	-64	-125

Cubing preserves the sign: positives stay positive, negatives stay negative. Squaring does not.

2. Find the volume of the cube shown.



$$V = 3^3 = 3 \times 3 \times 3 = 27 \text{ cubic meters}$$

A student who answers 9 has stopped after squaring.

3. A cube-shaped box has a volume of 8 cubic centimeters. How long is one edge?

$$2 \times 2 \times 2 = 8, \text{ so each edge is 2 centimeters.}$$

For questions 4 through 7, find the value.

4. $\sqrt[3]{1}$
1

5. $\sqrt[3]{-27}$

-3 $(-3) \times (-3) \times (-3) = -27$

A cube root of a negative number exists. A square root of one does not.

6. $\sqrt[3]{64}$
4

7. $\sqrt[3]{-125}$
 -5

For questions 8 through 10, find every real solution of the equation.

8. $x^3 = 27$

$x = 3$ one solution, not two
 $(-3)^3 = -27$, so -3 does not work.

9. $a^3 = -8$

$a = -2$

10. $w^3 = 1$

$w = 1$

Not ± 1 . $(-1)^3 = -1$.

11. Select ALL of the values below that make $x^3 = 125$ true.

☒ 5

☐ -5

☒ $\sqrt[3]{125}$

☐ $-\sqrt[3]{125}$

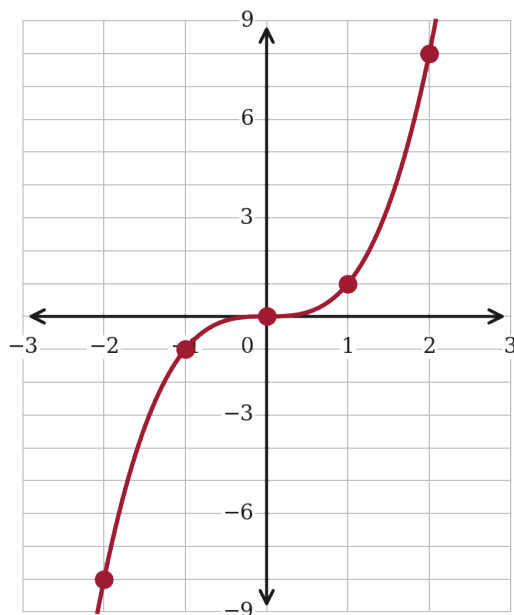
☐ 25

☐ $41\frac{2}{3}$

Two of the six. -5 and $-\sqrt[3]{125}$ both give -125 . 25 is 5^2 , and $41\frac{2}{3}$ is $125 \div 3$.

12. Complete the table of values for $y = x^3$. Then plot the five points on the grid.

x	Substitute into $y = x^3$	Coordinates (x, y)
-2	$y = (-2)^3 = -8$	$(-2, -8)$
-1	$y = (-1)^3 = -1$	$(-1, -1)$
0	$y = (0)^3 = 0$	$(0, 0)$
1	$y = (1)^3 = 1$	$(1, 1)$
2	$y = (2)^3 = 8$	$(2, 8)$



The graph passes through the origin and crosses the x-axis there. The graph of $y = x^2$ touches the x-axis at the origin and turns back up.

13. The equation $x^2 = 36$ has two solutions, but the equation $x^3 = 27$ has only one. Explain why.

$x^2 = 36$ has two solutions because $6 \times 6 = 36$ and $(-6) \times (-6) = 36$.

$x^3 = 27$ has one solution because $3 \times 3 \times 3 = 27$, but $(-3) \times (-3) \times (-3) = -27$.

Two negative factors give a positive; three give a negative.

The point needs the count of negative factors.

STRETCH. The two questions below use cubes larger than the Grade 8 FAST will ask for. The method is identical — only the arithmetic is bigger.

14. $\sqrt[3]{343}$

7 $7 \times 7 \times 7 = 343$

15. $m^3 = 512$

m = 8 $8 \times 8 \times 8 = 512$

Still one solution.